



Fourier Theory & Fourier Transform in Digital Image Processing

PRESENTED BY : **224**

1. What is Fourier Theory?

- **Definition** : Fourier Theory is a mathematical framework that allows us to represent complicated signals or images using simple sine and cosine waves.
- **Fourier theory was created by Joseph Fourier (1807–1822)**, a French mathematician and physicist.
- Fourier created the theory to represent complex signals using simple sine and cosine waves
- Even if an image looks complicated, behind the scenes it can be represented using repeating wave patterns.

2. Fourier Transform

- The Fourier Transform is the mathematical tool that converts the image from the spatial domain to the frequency domain.
- In Simple words :

1 Spatial Domain = Pixels

2 Frequency Domain = Frequencies

- A grayscale image contains pixel intensities ranging from **0 (black) to 255 (white)**.
- **Applications :**
 - Signal Processing
 - Communications
 - Engineering and Physics

3. Why Frequency Domain?

- A representation of a signal in terms of **its frequency components** instead of its original form.
- Usually, we work with images in the spatial domain, meaning we look directly at pixel values. But some operations become much easier if we convert the image into the frequency domain.

- In the frequency domain:

1

Repeating
patterns
become visible

2

Noise becomes
easier to
identify

3

Filtering becomes
more powerful
and precise

- This is why we use the frequency domain — it allows us to see and manipulate information that is hidden in the spatial domain.

4. Spatial Domain vs Frequency Domain

Spatial Domain

- You work **directly on the image's pixels**.
- You change the picture itself → like adjusting brightness or blurring by touching pixels.
- **Examples:** Brightness, contrast, blur, sharpen, edge detection.

Frequency Domain

- You **don't touch the pixels directly**.
- First, the image is converted into **frequencies** (using FFT).
- Then you change the **high frequencies** (details/edges) or **low frequencies** (smooth areas).
- **Examples:** Noise removal, image compression (JPEG), advanced filtering.

5. 2D Fourier Transform

- 2D Fourier Transform extends the Fourier Transform to two-dimensional signals, such as images.
- Converts a 2D spatial signal (image) into its **frequency components** in both horizontal and vertical directions.
- The result is a frequency spectrum which has two parts **Magnitude and Phase**.
- This formula is :

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)}$$

6. Continuous Fourier Transform(CFT)

- The **Continuous Fourier Transform** converts a **time-domain signal** into a **frequency-domain**.
- It essentially decomposes a signal into its sinusoidal components.
- Formula:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

7. Discrete Fourier Transform(DFT)

- DFT converts a **discrete-time signal** (finite number of samples) into its **frequency components**.
- Similar to Fourier Transform, but works with **digital signals**.
- Formula:

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn}$$

- For digital images, we use the Discrete Fourier Transform or DFT. But DFT is slow, especially for large images. So we use FFT.
- Computational complexity: $O(N^2)$

8. Fast Fourier Transform(FFT)

- FFT is an **efficient algorithm** to compute the Discrete Fourier Transform (DFT) quickly.
- DFT is slow, especially for large images. So in real systems, we use FFT(Fast Fourier Transform).
- FFT performs the same process but much faster, reducing time from **$O(n^2)$** to **$O(n \log n)$** .
- **Applications :**
 - Real-time signal processing (audio, video)
 - Image processing and compression
 - Communications and spectral analysis

9. Magnitude and Phase Spectrum

➤ When we take the Fourier Transform of an image we get :

1 Magnitude Spectrum

- shows how strong each frequency is
- **strength (amplitude)**

2 Phase Spectrum

- contains the actual structure of the image
- Shows Phase shift of each frequency component
- Tell how each frequency is shifted

➤ The phase contains most of the important information.

➤ Importance :

If we keep the phase and remove magnitude, the image is still recognizable. But if we keep only the magnitude and remove phase, the image becomes unrecognizable.

10. Frequency Domain Filters

➤ In frequency domain we can apply different types of filters :

1 Low-Pass filter

- Remove high frequencies
- Smooth the image

2 High-Pass filter

- Remove low frequencies
- Highlight edges

3 Band-Pass/Notch filter

- Remove or allow specific frequency ranges
- Remove noise

- These are extremely power full for Operations like image smoothing , sharpening or noise reduction
- Central Bright area is low frequency.
- Side Bright area is high frequency.

11. Applications

➤ The Fourier Transform has many applications, including :

- 1 **Edge Detection**
- 2 **Noise Removal**
- 3 **Image Smoothing**
- 4 **JPEG Compression**
- 5 **Texture and Pattern analysis**

➤ In fact, almost every advanced image processing system uses Fourier concepts.

Thank You